

Timothy Markfeld

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SUMMARY

Senior Software Engineer focused on distributed systems, backend platforms, and AI-powered automation products. Experience designing and operating high-throughput systems across healthcare and SaaS environments using .NET, AWS, GraphQL, Kafka, TypeScript, and cloud-native infrastructure.

EXPERIENCE

Senior Software Engineer

Apr. 2025 – Present

FirmPilot

Nashville, TN

- Built and owned fault-tolerant event-based worker systems leveraging AWS SQS/SNS, ECS, RDS, and Parameter Store, with retry logic and observability supporting content, social media, and backlink automation workflows.
- Designed and implemented secure GraphQL APIs and React/Next.js TypeScript frontends with optimized database access patterns maintaining p99 query latency under 25ms across multi-tenant workloads.
- Engineered end-to-end AI-driven content pipelines integrating LLMs, fact-checking, media generation, and automated WordPress publishing.
- Led architecture cleanup and large-scale refactors, introducing CQRS patterns to improve long-term system maintainability.
- Reviewed and merged 1,500+ PRs across backend, frontend, and infrastructure systems while conducting daily code reviews and mentoring engineers on architecture and best practices.
- Active on-call responder with 97.8% incident acknowledgment rate, 13s MTTA, and 22min MTTR across 45+ production incidents, with zero timeout escalations.

Data Integration Engineer → Data Integration Engineer II

Sep. 2023 – Apr. 2025

HCA Healthcare

Nashville, TN

- Supported and modernized a healthcare integration platform consisting of 200+ HIPAA-regulated .NET microservices processing over 1M daily events.
- Integrated with external EHR systems using FHIR APIs, HL7 messaging, and IHE interoperability standards while supporting IBM MQ and Kafka-based event processing.
- Helped architect and lead migration of legacy integration services to Apache NiFi, including infrastructure provisioning with Terraform, Azure AD integration, and deployment on Google Cloud Platform.
- Implemented Dynatrace monitoring, alerting, and automated failover workflows while mentoring engineers on interoperability and platform architecture.

EDUCATION

The University of Alabama

Tuscaloosa, AL

M.S. Management Information Systems, GPA: 4.0/4.0

Jan. 2021 – May 2023

B.S. Management Information Systems, GPA: 3.94/4.0

Sep. 2019 – May 2023

PROJECTS

HL7Kit / HL7Kit.Fhir | .NET, HL7, FHIR

github.com/tcmarkfeld/HL7Kit

- Strongly typed HL7 v2 parser with full segment coverage, dynamic delimiter handling, Z-segment support, and FHIR R4 conversion for .NET. Published on NuGet.

Conductor | .NET, Terraform, MassTransit

github.com/tcmarkfeld/Conductor

- CLI that generates least-privilege IAM policies and Terraform resources from MassTransit/SQS/SNS config for .NET teams deploying to AWS. Published on NuGet.

Corolla Ice Delivery | React Native, React, SQL, Node.js

- Mobile app that optimizes delivery routes, reducing delivery time by up to 3 hours/day.
- Owned the full development lifecycle from initial design through deployment to the iOS App Store and Google Play.

TECHNICAL SKILLS

Languages: C#, Java, SQL (PostgreSQL, SQL Server), Python, JavaScript/TypeScript

Infrastructure: Terraform, Docker

Frameworks: .NET, React, Next.js, Node.js, Entity Framework, React Native

Technologies: AWS (ECS, ECR, SQS, SNS, RDS, Parameter Store, Secrets Manager), GCP, Kafka, GraphQL, MassTransit, Docker, Datadog, Dynatrace